Name : vizanth V

Reg No :192512351

Course name : Advanced Wireless Commincation

Course code : ECA5610

MATLAB – Analysis of 5G NR Frame Structure and Subcarrier Spacing

Aim:

To analyze the 5G NR frame structure by studying subcarrier spacing, slot duration, OFDM symbols and simplified downlink/uplink resource assignments using MATLAB, and to relate the radio resources to gNodeB functions and RRC operation.

Objectives:

1. To generate or inspect a simplified 5G NR resource grid.
2. To compare selected subcarrier spacings and their effect on symbol or slot duration.
3. To identify basic downlink and uplink resource assignments in the model.
4. To relate the radio resources to gNodeB functions, RAN organization and RRC's role

Requirements:

- MATLAB

Program :Objective 1

```
clc; clear; close all;

Nsc = 24;                % Simplified number of subcarriers
Nsym = 14;               % OFDM symbols per slot

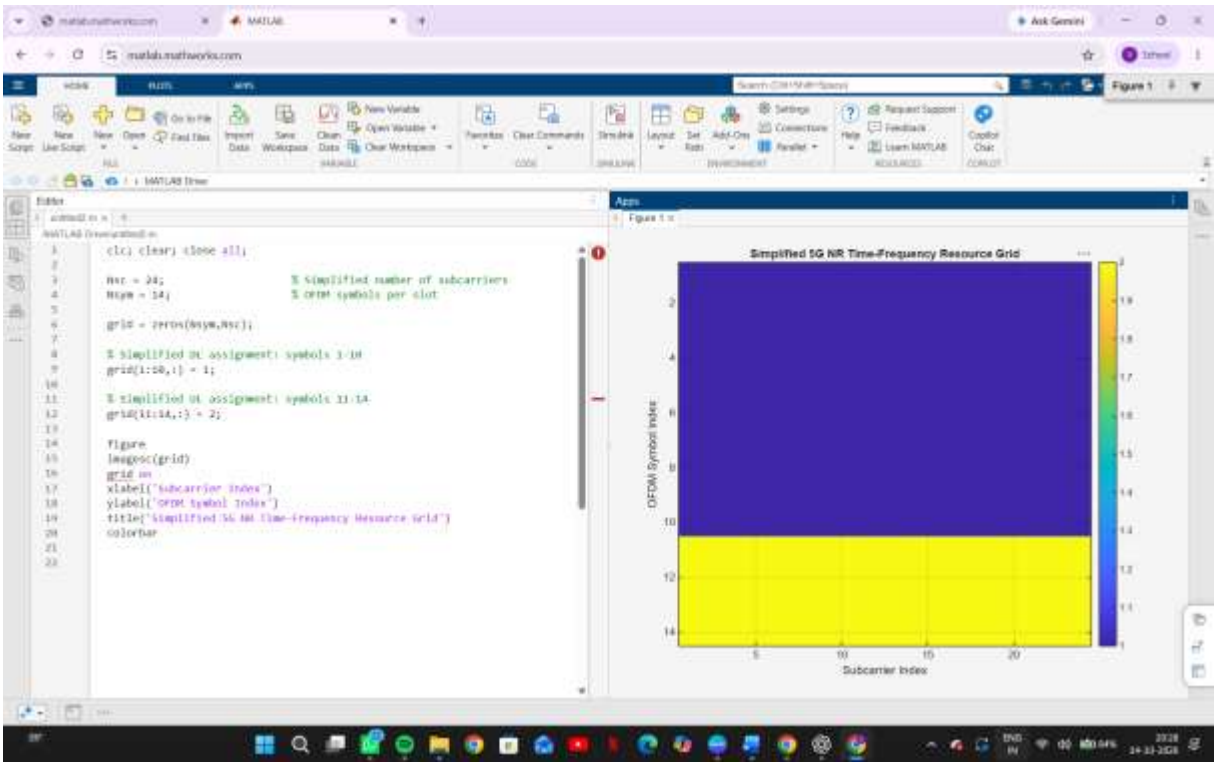
grid = zeros(Nsym,Nsc);
```

```
% Simplified DL assignment: symbols 1-10
grid(1:10,:) = 1;

% Simplified UL assignment: symbols 11-14
grid(11:14,:) = 2;

figure
imagesc(grid)
grid on
xlabel('Subcarrier Index')
ylabel('OFDM Symbol Index')
title('Simplified 5G NR Time-Frequency Resource Grid')
colorbar
```

Output:



Program :Objective 2

```
clc; clear; close all;

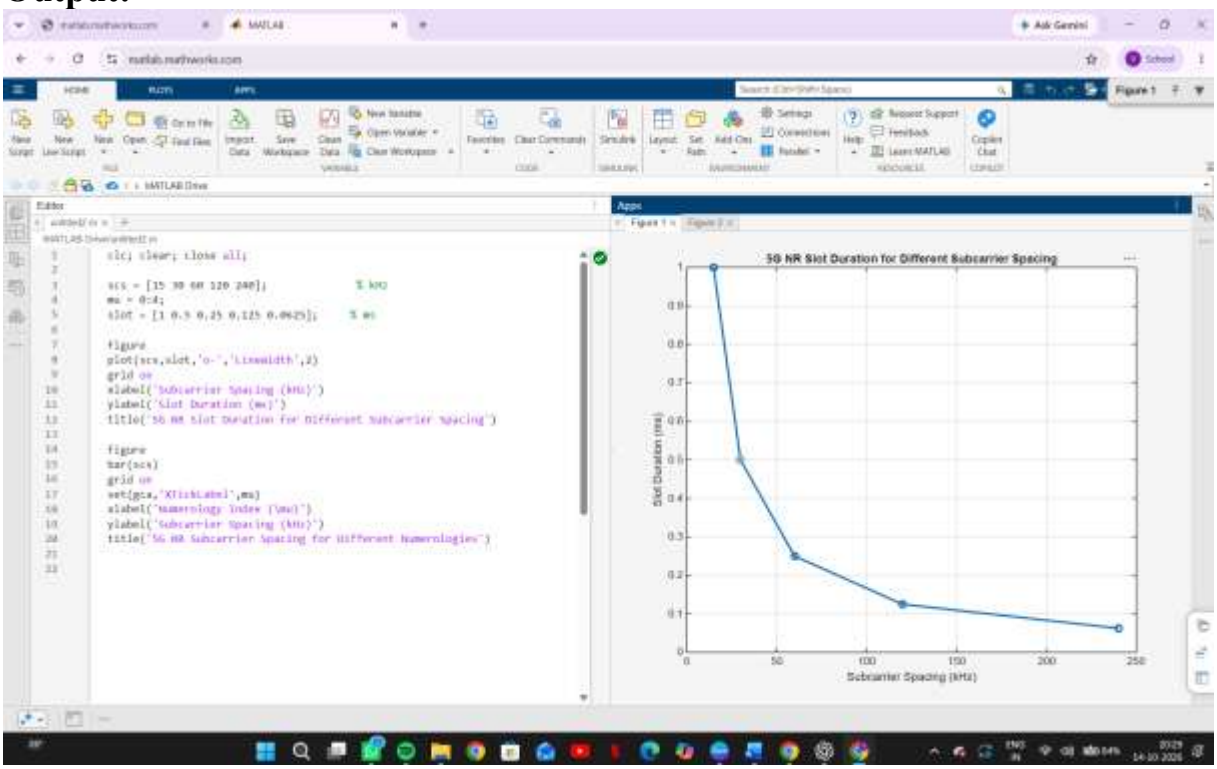
scs = [15 30 60 120 240];              % kHz
mu = 0:4;
slot = [1 0.5 0.25 0.125 0.0625];     % ms

figure
```

```
plot(scs,slot,'o-','LineWidth',2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')

figure
bar(scs)
grid on
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```

Output:



Program :Objective 3

```
clc; clear; close all;

symbols = 1:14;
DL = [1 1 1 1 1 1 1 1 1 1 0 0 0 0];
UL = [0 0 0 0 0 0 0 0 0 0 1 1 1 1];

figure
plot(symbols,DL,'o-','LineWidth',2); hold on
plot(symbols,UL,'s-','LineWidth',2)
grid on
```

```
xlabel('OFDM Symbol Index')
ylabel('Allocation Status')
title('Simplified Downlink and Uplink Assignment')
legend('Downlink','Uplink')
```

Output:

